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DATA ANALYTICS WITH TABLEAU ASSIGNMENT

1.AREA CHART

Drag Order Date to Columns shelf

- 1 find **Order Date** under **Dimensions**.
2. Drag **Order Date** to the Columns shelf.
 - Tableau automatically shows **YEAR(Order Date)** in the Columns shelf.
3. Click the drop-down arrow on **YEAR(Order Date)**. Select Month (green Month option).
 - The Columns shelf changes to **MONTH(Order Date)**.

Drag Sales to Rows

- 1.Find **Sales** under Measures.
- 2.Drag **Sales** to the **Rows** shelf.
 - Tableau automatically changes it to **SUM(Sales)**.

Change Chart Type to Area

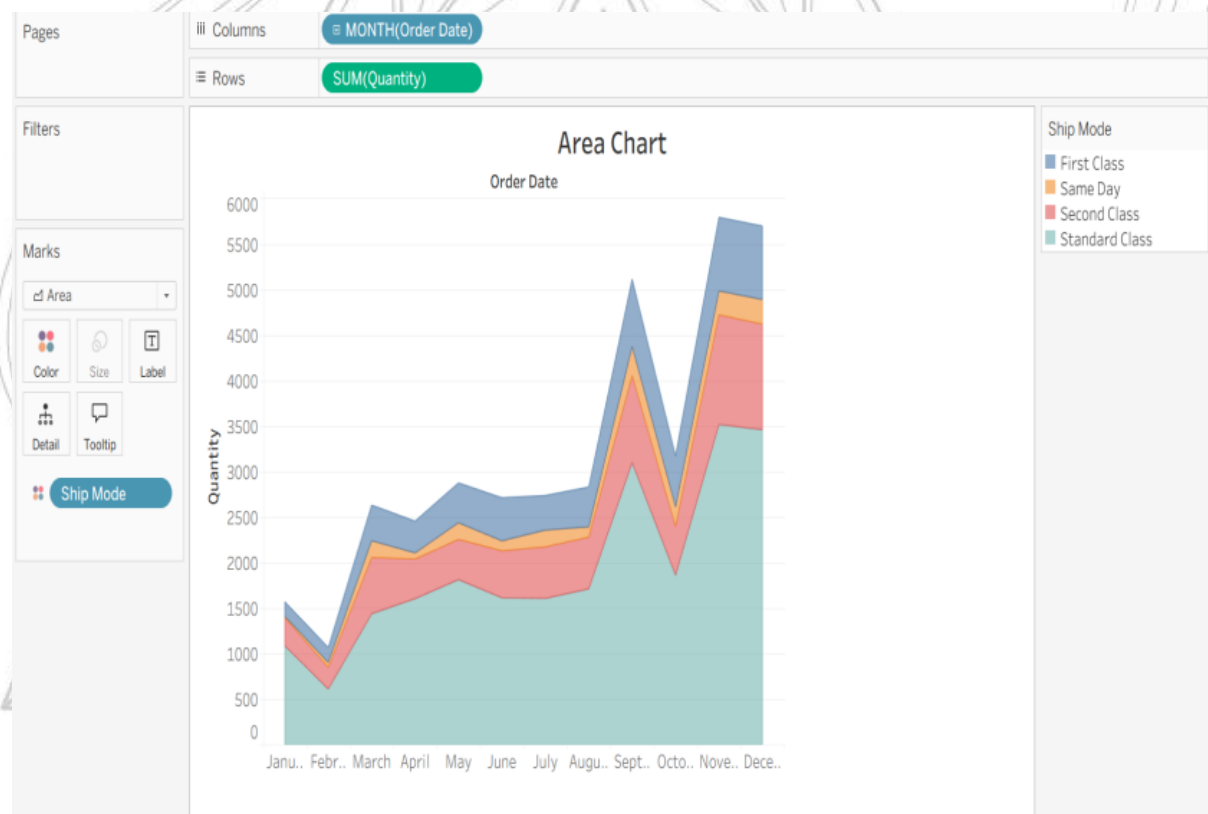
- 1.In the Marks card, click the drop-down menu that says **Automatic**.
- 2.Select **Area**.

Drag Sub-Category to Color

1. Find **Sub-Category** under Dimensions.

2. Drag **Sub-Category** to **Color** in the Marks card.

OUTPUT:



2. BAR CHART

Drag Region to Rows

1. In the left panel, select **Region** under Dimensions.

2. Drag **Region** to the **Rows shelf**.

- Tableau shows all regions (East, West, South, Central) on the left side.
3. Click the **drop-down arrow** on Region in the Rows shelf.
 4. Select Filter.
 5. Check only West.
 6. Click OK

Drag Sales to Rows

1. Select **Sales** under **Measures**.
2. Drag **Sales** to the Rows shelf, **next to Region**.
 - Tableau automatically changes it to SUM(Sales).

Change Chart Type to Bar

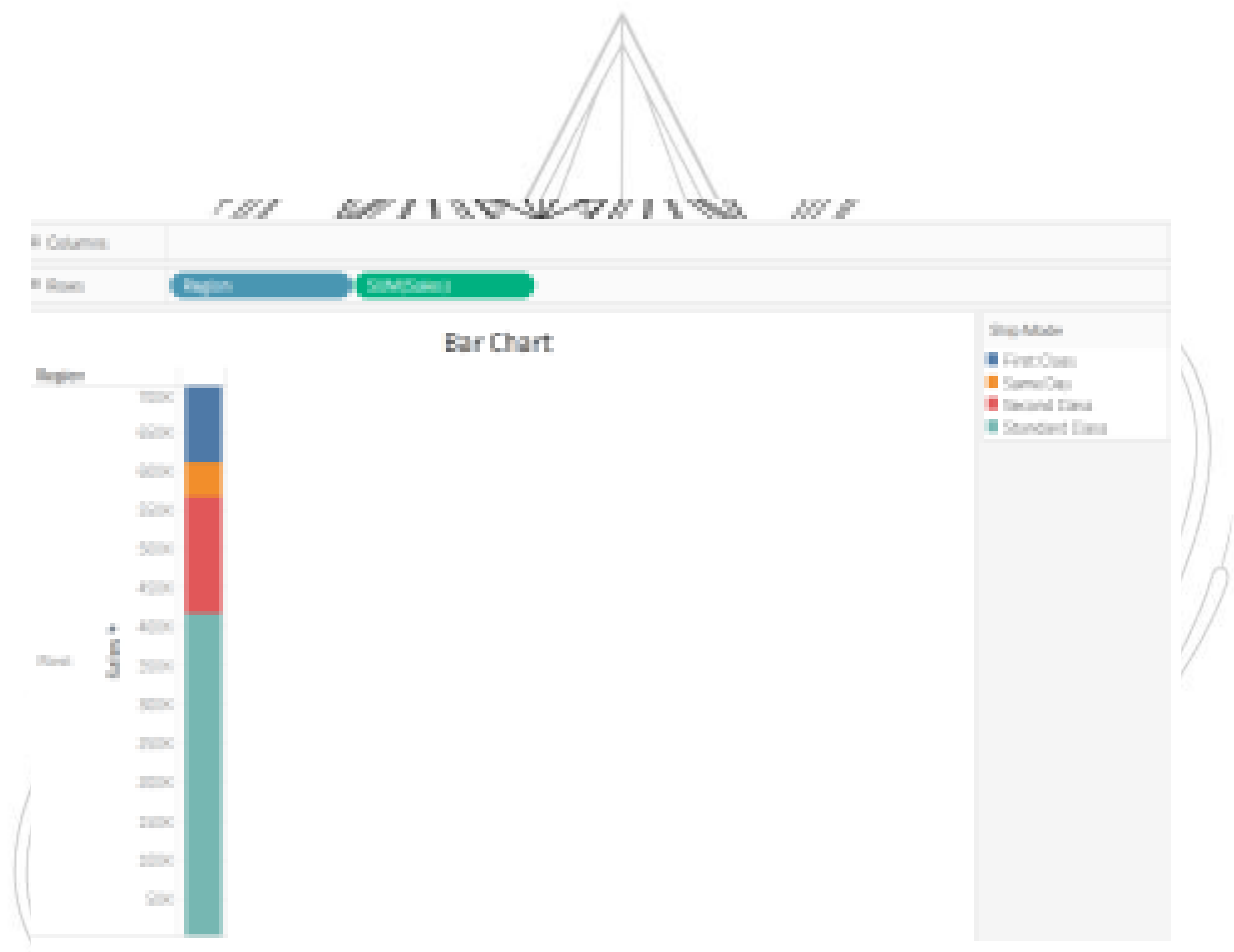
- In the **Marks card**, click the **drop-down** menu.
- Select **Bar**.

Drag Ship Mode to Color

1. Select **Ship Mode** under **Dimensions**.
2. Drag **Ship Mode** to **Color** in the Marks card.
 - The Bar Chart forms with different colours..

OUTPUT:

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3. BOX PLOT

Drag Discount to columns shelf

1. Drag **Discount** to the **Columns shelf**.

- An Horizontal axis appears

Drag Segment to Rows shelf.

2. Drag **Segment** to the **Rows shelf**.

- The segments appear on the left side.

3. Drag **Region** to the **left side of Segment** in the Rows.



Click Show Me

4. Click **Show Me** on the top-right side.

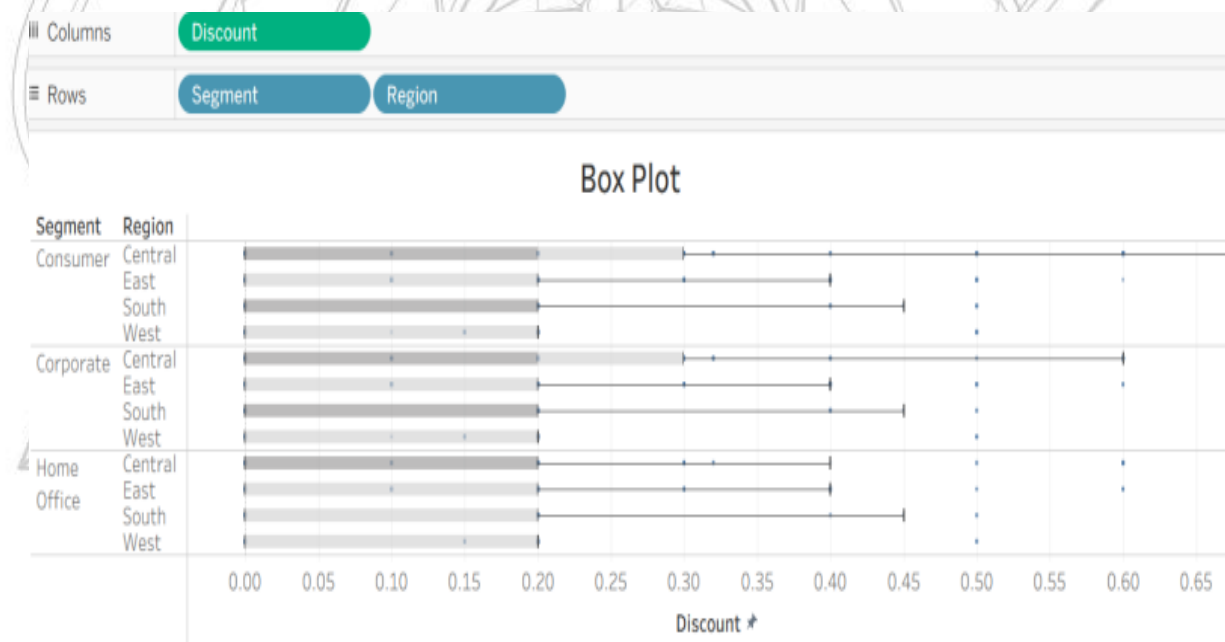
5. Click **Box-and-Whisker plot** icon.

Swapping

6. Click Swap Axes (Rows and Column) button.

7. The Box Plot is formed.

OUTPUT:



4. Gantt Chart

Drag Sub-Category to Rows

1. In the left panel, find **Sub-Category** under **Dimensions**.
2. Drag **Sub-Category** to the **Rows shelf**.
 - Tableau shows all sub-category names on the left side.

Drag Ship Mode to Rows

3. Find **Ship Mode** under **Dimensions**.
4. Drag **Ship Mode** to the Rows shelf, next to Sub-Category.
 - Each sub-category is divided into different ship modes.

Drag Order Date to Columns

5. Select **Order Date** under **Dimensions**.
6. Drag **Order Date** to the **Columns shelf**.
 - Tableau automatically shows **YEAR(Order Date)**.

Change Year to Month

7. Click the drop-down arrow on **YEAR(Order Date)**.
8. Select Month (green Month option).
 - The Columns shelf changes to **MONTH(Order Date)**.

Change Mark Type to Gantt Bar

9. In the **Marks card**, click the drop-down menu that says Automatic.
10. Select **Gantt Bar**

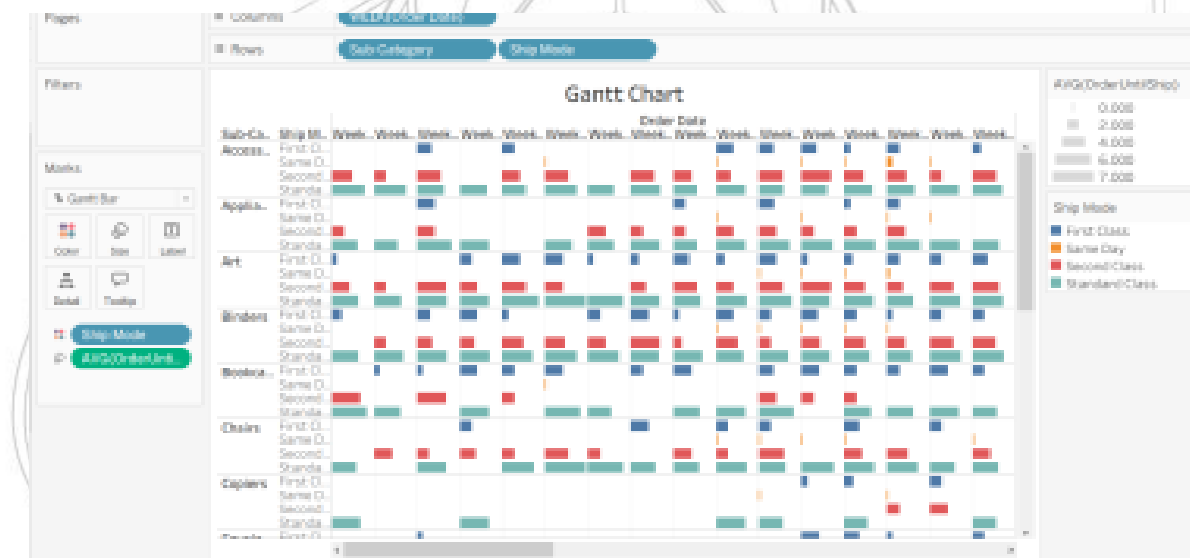
Drag Ship Mode to Color

11. Drag Ship Mode to Color in the Marks card.

Drag AVG(Order ID) to Size

- Drag **Order ID** to the **Size** option in the Marks card

OUTPUT:



5.HEAT MAP

Drag Segment to Columns

1. Select **Segment** under **Dimensions**.
2. Drag **Segment** to the **Columns** shelf.
 - Tableau shows segment names (Consumer, Corporate, Home Office) at the top.

Drag Region to Rows

3. Select **Region** under **Dimensions**.
4. Drag **Region** to the **Rows** shelf.

- Tableau shows regions (Central, East, South, West) on the left side.

Drag Sub-Category to Rows

5. Select **Sub-Category** under **Dimensions**.
6. Drag **Sub-Category** to the Rows shelf, next to **Region**.
 - Each region is divided into different sub-categories.

Change Mark Type to Square

7. In the **Marks card**, click the drop-down menu that says Automatic.
8. Select **Square**.
 - Small squares appear in the worksheet.

Drag Profit to Color

9. Find **Profit** under **Measures**.
10. Drag **Profit** to **Color** in the Marks card.

Drag Sales to Size

11. Find **Sales** under **Measures**.
12. Drag **Sales** to **Size** in the Marks card.
 - Tableau automatically changes it to SUM(Sales).

Adjust Size (Optional)

13. Click **Size** in the Marks card.
 - Move the slider to make the squares bigger

OUTPUT:



6. HISTOGRAM

Drag Quantity to Columns

1. From **Measures**, drag **Quantity**
2. Drop it on **Columns Shelf**

Create Histogram Using Show Me

3. Click **Show Me**
4. Select **Histogram**

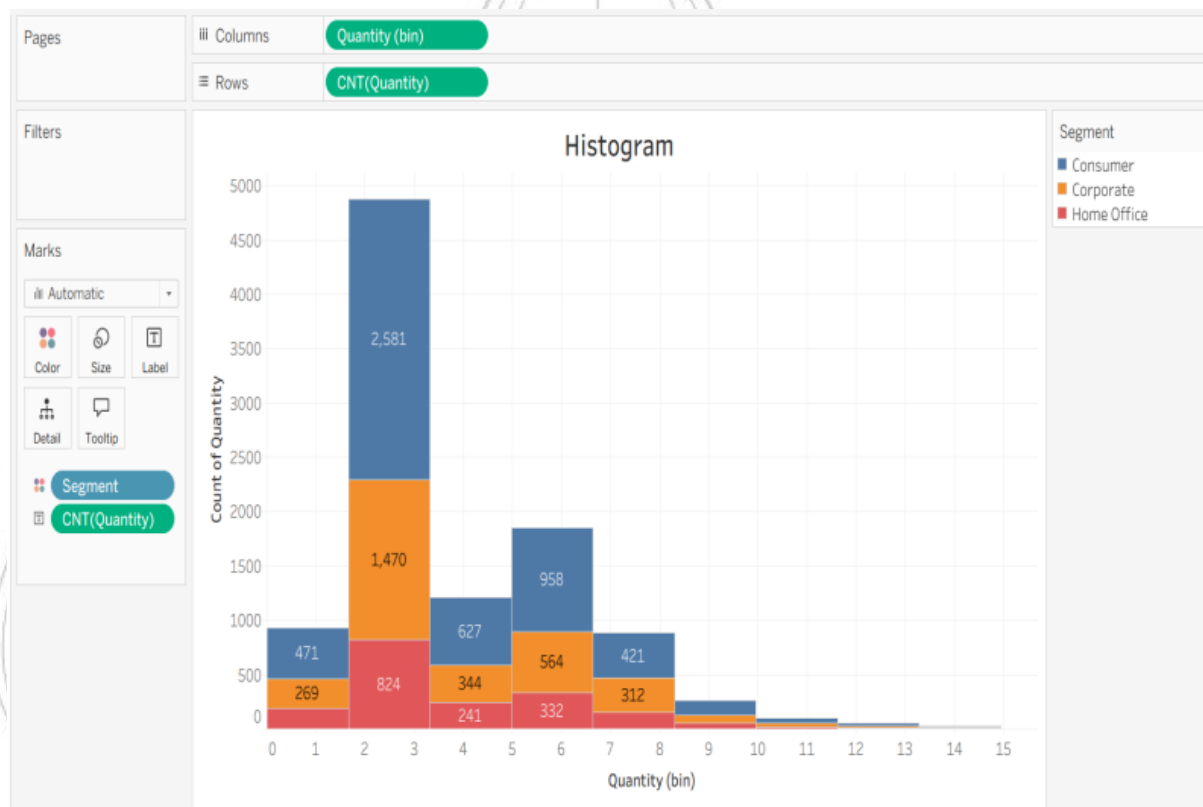
Add Segment to Color

5. Drag **Segment** from **Dimensions**
6. Drop it on **Color** in Marks card Now bars are formed.

Copy Count to Label

7. Hold Ctrl

8. Drag **CNT(Quantity)** from **Rows shelf** to Label on Marks card (Using Ctrl copies field.)



7. LINE CHART

Drag Ship Date to Columns

1. From **Dimensions**, drag **Ship Date**
2. Drop it on **Columns Shelf**
3. **Right-click** on Ship Date in Columns and change it to **Month** (the continuous format showing Month Year)

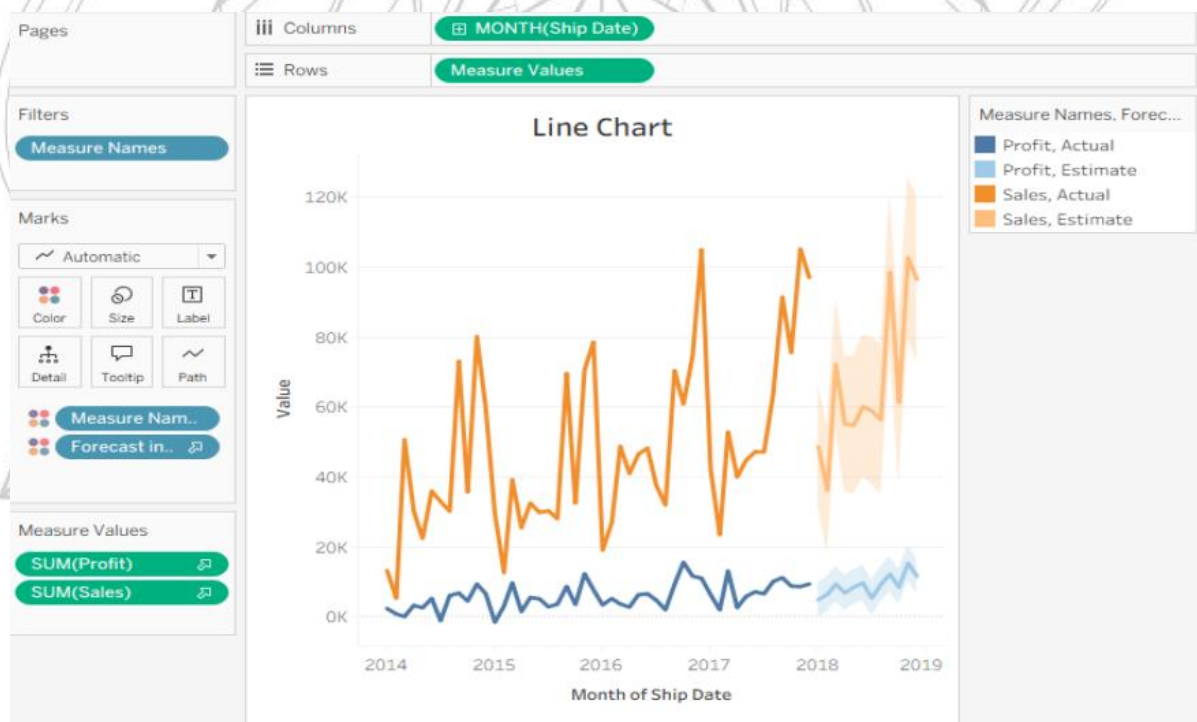
Drag Profit and Sales to Rows

4. From **Measures**, drag **Profit** and drop it on **Rows Shelf**
5. Drag **Sales** and drop it **right next to Profit** on Rows Shelf

- Creates a Measure Values card showing **SUM(Profit)** and **SUM(Sales)**
- Adds Measure Names to the Filters shelf and Color in Marks card

Create Forecast

7. Go to the **Analysis** menu at the top
8. Navigate to **Forecast** > Show Forecast
9. Drag **Forecast** to Workspace and **Drop it**.



8. PACKED BUBBLE CHART

Drag Category, Region, and Sales to Columns and Rows

1. From **Dimensions**, drag **Category** and **Region** to the **view** or **Marks** card.
2. From **Measures**, drag **Sales** to the view.

Create Packed Bubble Chart

3. Click **Show Me**

4. Select **Packed Bubbles** icon.

- Clears Columns and Rows shelves (keeps them blank).
- Changes the Marks card type to Circle.
- Places SUM(Sales) on Size.
- Places Category and Region on Detail / Label.

:Add Profit to Color

5. From **Measures**, drag **Profit** and drop it onto **Color** in the Marks card.

- orange (losses) , dark blue (high profit) as shown in the SUM(Profit) legend.

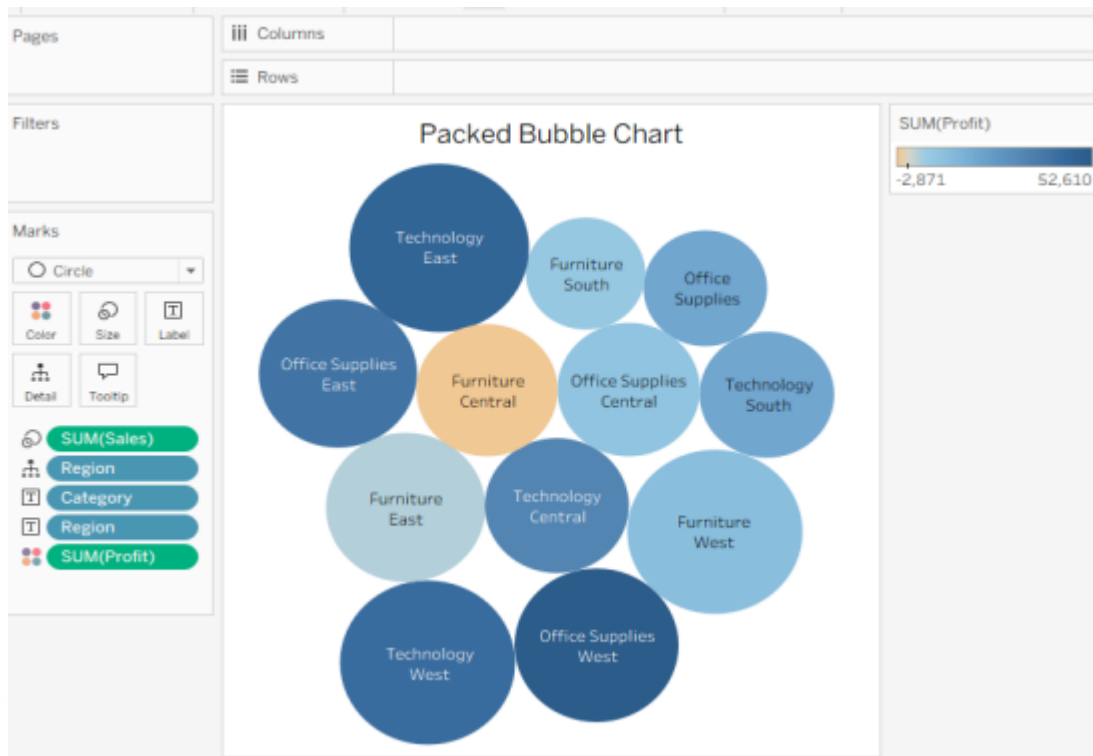
Add Labels to Bubbles

6. Verify Category and Region are placed on the Label icon in the Marks card.

7. If missing, drag them to Label so the text appears inside each bubble.

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9. PIE CHART

Drag Category and Profit to the view

1. From **Dimensions**, drag **Category**
2. From Measures, drag Profit and drop it into the view

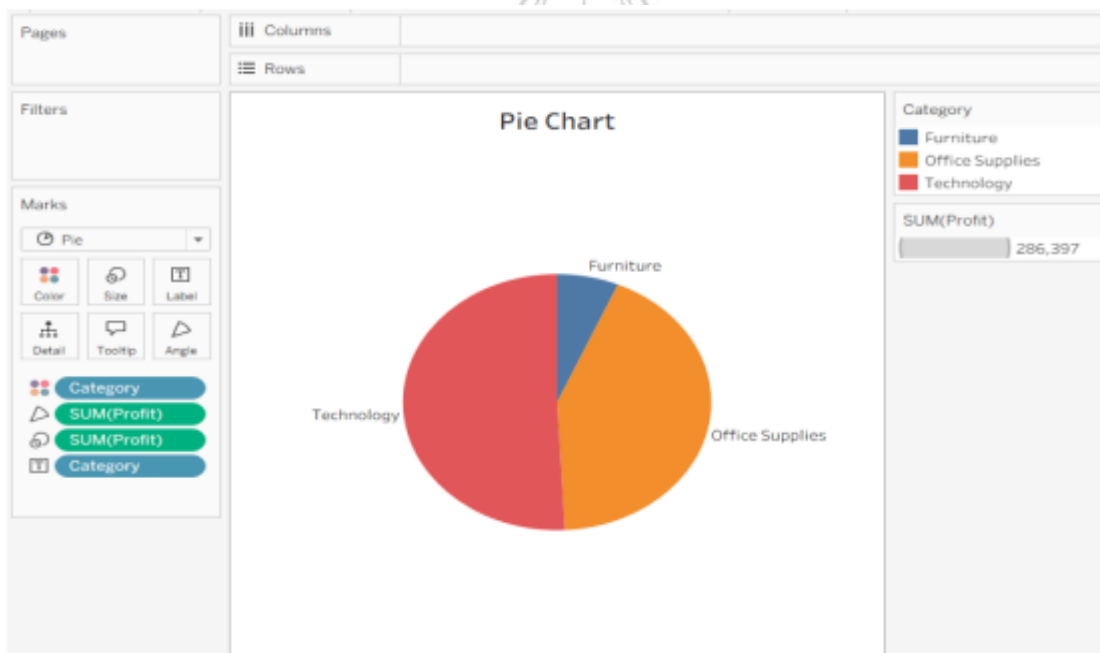
Create Pie Chart Using Marks card

3. Click Pie in Marks card.
4. Select Pie Charts, Changes the Marks card type to Pie
 - Places Category on Color
 - Places SUM(Profit) on Angle

Add Labels to Slices

1. Drag **Category** from **Dimensions** and drop it on **Label** in Marks card.

2. Hold Ctrl, drag **SUM(Profit)** from Angle/Marks card and drop it on **Label** if you want to see values



10. SCATTER PLOT

Drag Sales and Profit to Columns and Rows

1. From **Measures**, drag **Sales** and drop it on the **Columns** shelf.
2. From **Measures**, drag **Profit** and drop it on the **Rows** shelf.

Add Detail to Disaggregate Data

3. From **Dimensions**, drag **Region** and drop it on **Detail** in Marks card.

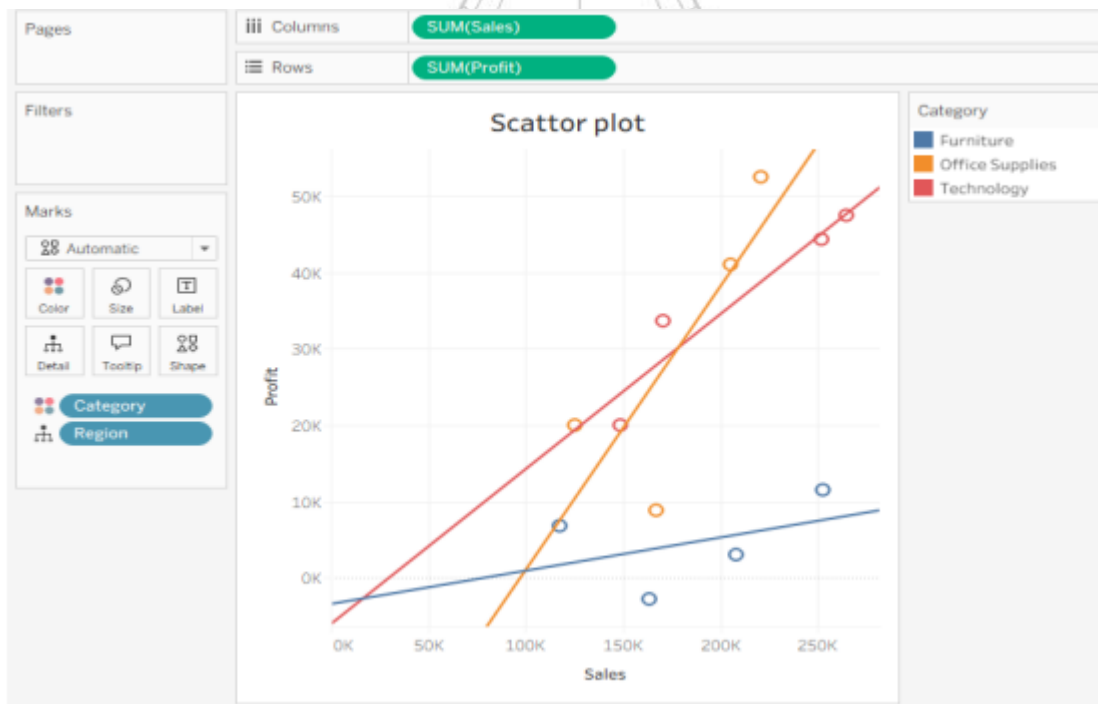
Add Category to Color

4. From **Dimensions**, drag **Category** and drop it on **Color** in the Marks card.

Add Trend Lines

5. Go to the **Analytics** pane.

6. Drag and drop Trend Line onto the view, selecting Linear.



11. TABLE WITH PERCENTAGE

Drag Order Date to Columns

1. From **Dimensions**, drag **Order Date**
2. Drop it on **Columns Shelf**
 - Sets it to YEAR(Order Date) Step 2: Drag Region and Sub-Category to Rows

Drag Region to Rows shelf

3. From **Dimensions**, drag **Region** and drop it on **Rows Shelf**

4. Drag **Sub-Category** and drop it right next to Region on Rows Shelf

Add Sales to Text Label

5. From **Measures**, drag **Sales**

6. Drop it on **Text (or Label)** in the Marks card

- Displays SUM(Sales) values in the table grid
- Changes Marks card type to Text or Automatic

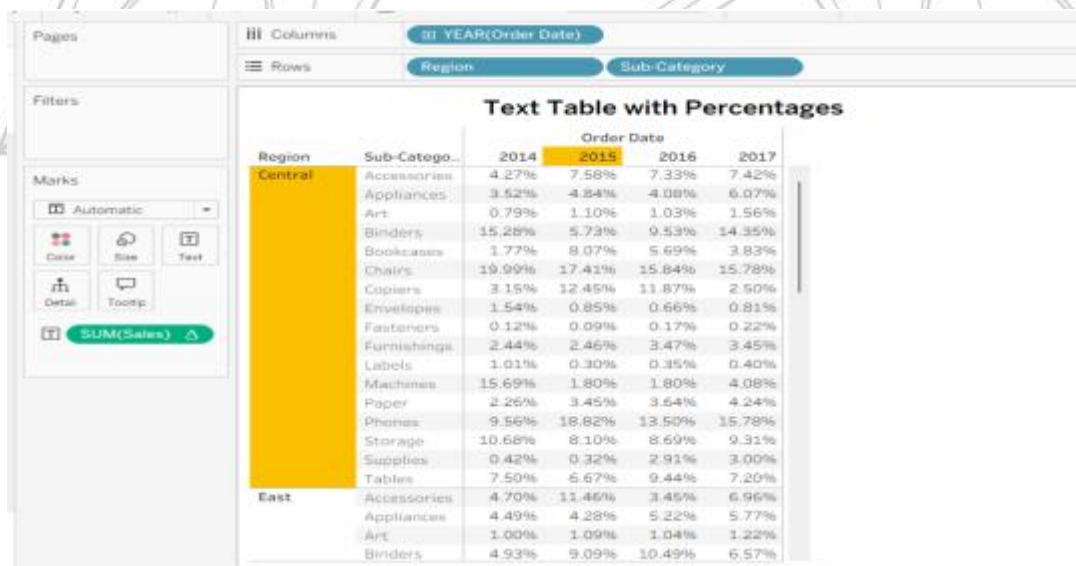
Convert Labels to Percent of Total

7. Right-click **SUM(Sales)** on the Marks card

8. Select: **Quick Table Calculation -> Percent of Total**

9. Right-click on it again, go to Compute Using, and select **Pane (down)** or your preferred direction to match the exact view percentages.

OUTPUT:



The screenshot shows the Tableau interface with a text table visualization titled "Text Table with Percentages". The table displays sales data as percentages of the total for each sub-category across four years (2014-2017), grouped by region (Central and East). The 'Central' region is highlighted in orange.

Region	Sub-Category	Order Date			
		2014	2015	2016	2017
Central	Accessories	4.27%	7.58%	7.33%	7.42%
	Appliances	3.52%	4.84%	4.08%	6.07%
	Art	0.79%	1.10%	1.03%	1.56%
	Binders	15.28%	5.73%	9.53%	14.35%
	Bookcases	1.77%	8.07%	5.69%	3.83%
	Chairs	19.99%	17.41%	15.84%	15.78%
	Copiers	3.15%	12.45%	11.87%	2.50%
	Envelopes	1.54%	0.85%	0.66%	0.81%
	Fasteners	0.12%	0.09%	0.17%	0.22%
	Furnishings	2.44%	2.46%	3.47%	3.45%
	Labels	1.01%	0.30%	0.35%	0.40%
	Machines	15.69%	1.80%	1.80%	4.08%
	Paper	2.26%	3.45%	3.54%	4.24%
	Phones	9.56%	18.82%	13.50%	15.78%
	Storage	10.68%	8.10%	8.69%	9.31%
East	Supplies	0.42%	0.32%	2.91%	3.00%
	Tables	7.50%	5.67%	9.44%	7.20%
	Accessories	4.70%	11.46%	3.45%	6.96%
	Appliances	4.49%	4.28%	5.22%	5.77%

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12. TREE MAP

Drag Sub-Category and Sales to the view

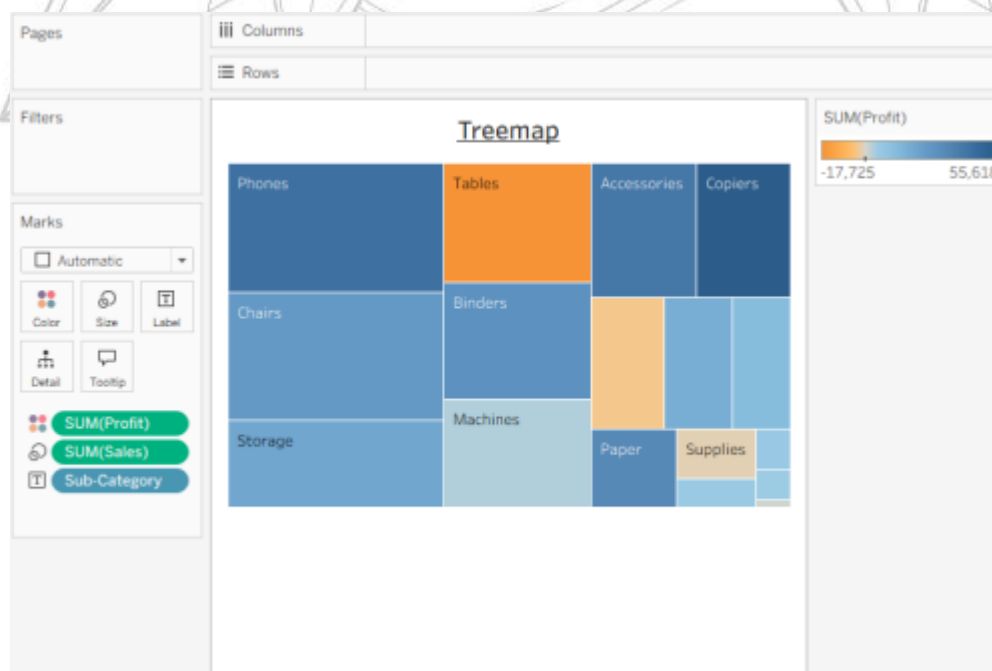
1. From **Dimensions**, drag **Sub-Category**
2. From **Measures**, drag **Sales** and drop it into the **view**

Create Treemap Using Show Me

3. Click **Show Me**
4. Select **Treemaps** icon Tableau Automatically Does:
 - Places SUM(Sales) on Size
 - Places Sub-Category on Label and Detail

Add Profit to Color

5. From Measures, drag Profit and drop it onto Color in the Marks card.
6. This creates the color gradient from orange (losses) to deep blue (high profit) as shown in the SUM(Profit) legend.



13. DONUT CHART

Create a Placeholder Field (Zero) and Drag to Rows

1. Create a **calculated field named zero with the formula 0.**
2. Drag **this SUM(zero)** field onto the Rows shelf twice.
Tableau Automatically Does:
 - Creates two separate Marks cards on the left named **SUM(zero)** and **SUM(zero) (2)**.
 - Generates two identical axis lines in the view.

Configure the First Pie Chart

3. Click on the first SUM(zero) Marks card to expand it.
4. Change the mark type from Automatic to Pie.
5. Drag **Region** from **Dimensions** and drop it on **Color**.
6. Drag **Sales** from **Measures** and drop it on **Angle**

Configure the Center Hole

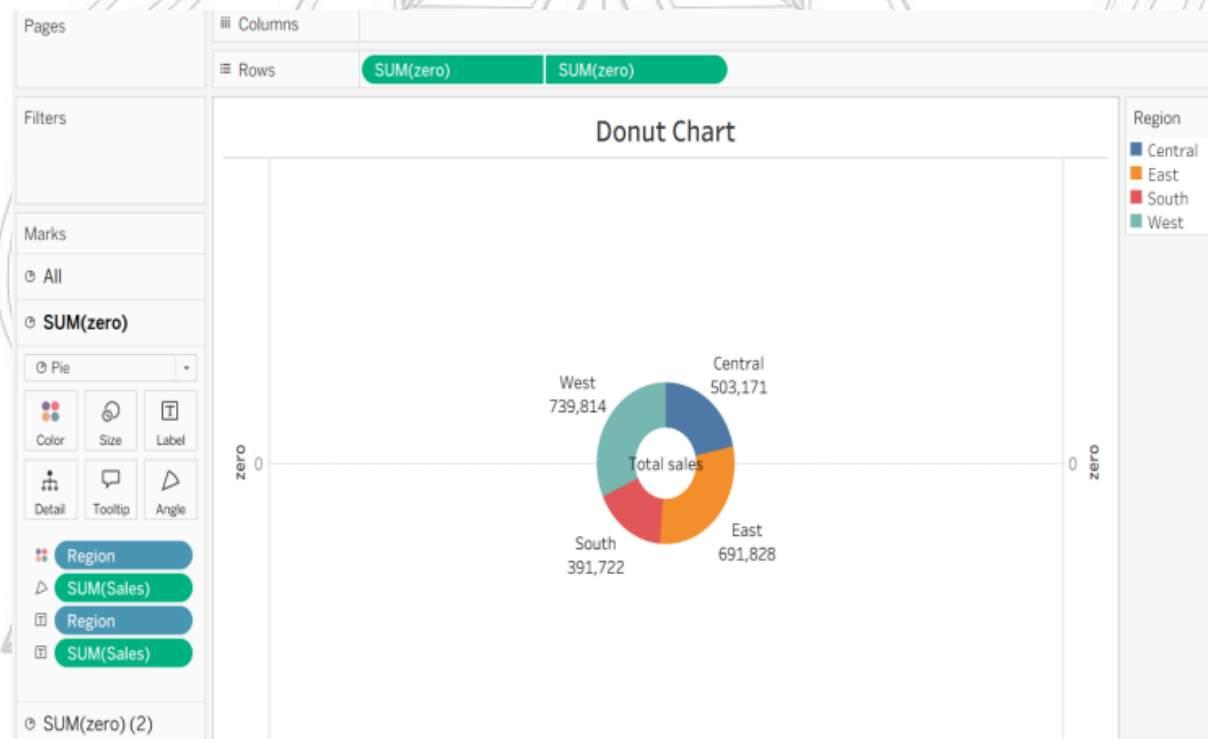
7. Click on the **second SUM(zero) (2)** Marks card to expand it.
8. Change the **mark type to Pie.**
9. select **Color**, and change it to **White**
10. Click on **Size** and make this **white circle** slightly smaller than the first pie chart.
11. Click on **Label** and type **"Total sales"** to display it inside the center hole.

Combine into a Donut Chart using Dual Axis

12. Right-click on the **second SUM(zero)** field in the Rows shelf.

13. Select **Dual Axis**. Tableau Automatically Does:

- Overlaps the white circle directly on top of the colored pie chart, creating a donut effect.



14. MAP CHART

Double-click State Field to Generate Map

1. From **Dimensions**, double-click on **State/Province** (or drag it to the center of the view). Automatically it shows
- Longitude (generated) on Columns

- Latitude (generated) on Rows
- Plots data points for each state on a map background.

Change Marks to Map (Filled Map)

2. Go to the **Marks** card dropdown.
 3. Change the chart type selection from Automatic/Circle to Map.
- Converts individual state points into filled geographic shapes.

Add Sales to Color

4. From **Measures**, drag **Sales** and drop it on **Color** in the Marks card.
 5. Creates a **color** gradient scale (purple) based on **SUM(Sales)**.
- Darker shades represent higher sales, and lighter shades represent lower sales.

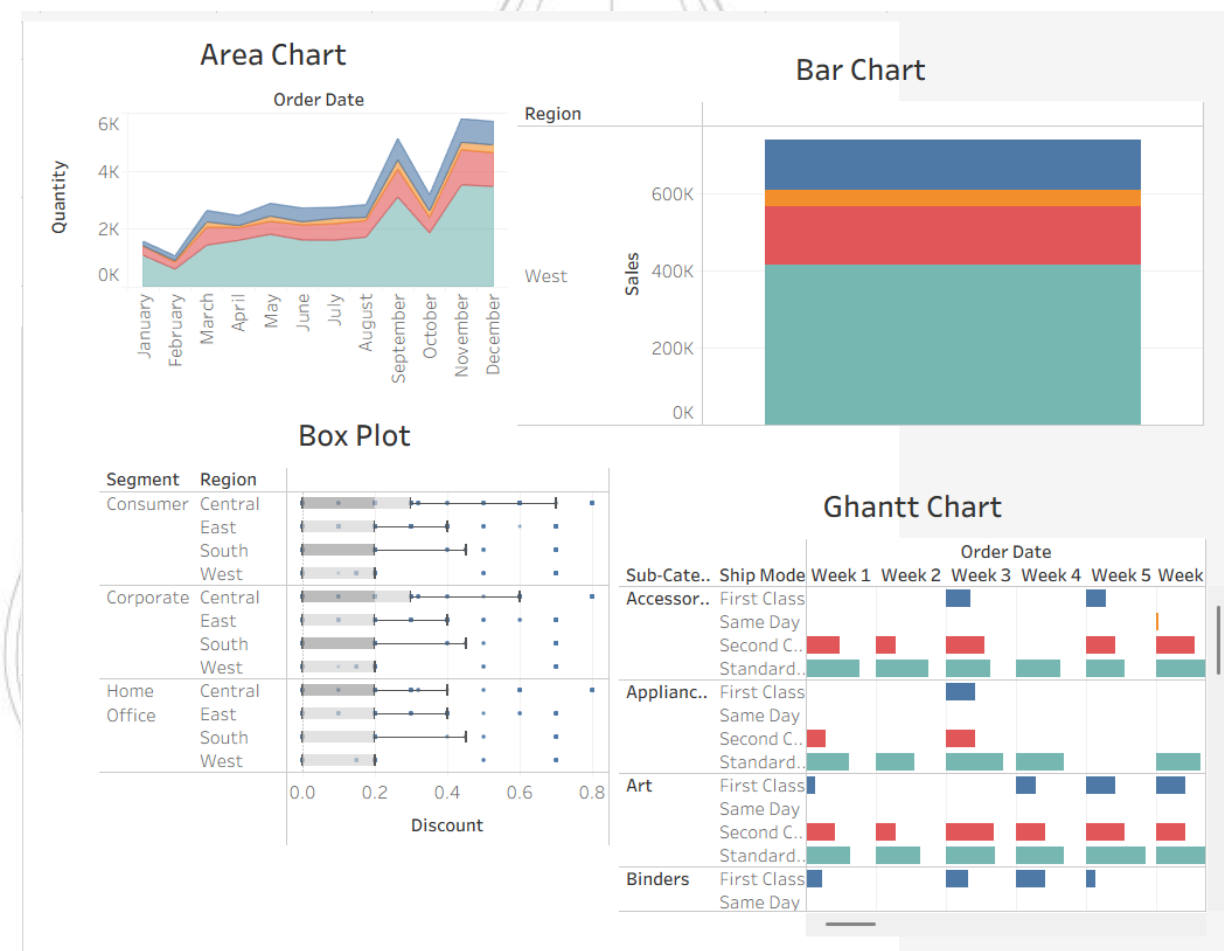
Add Labels to States

6. Drag **State/Province** from Dimensions and drop it on **Label** in the Marks card.
7. Drag **Sales** from **Measures** and drop it on **Label** as well to show the exact sales figures under the state names.

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DASHBOARD 01

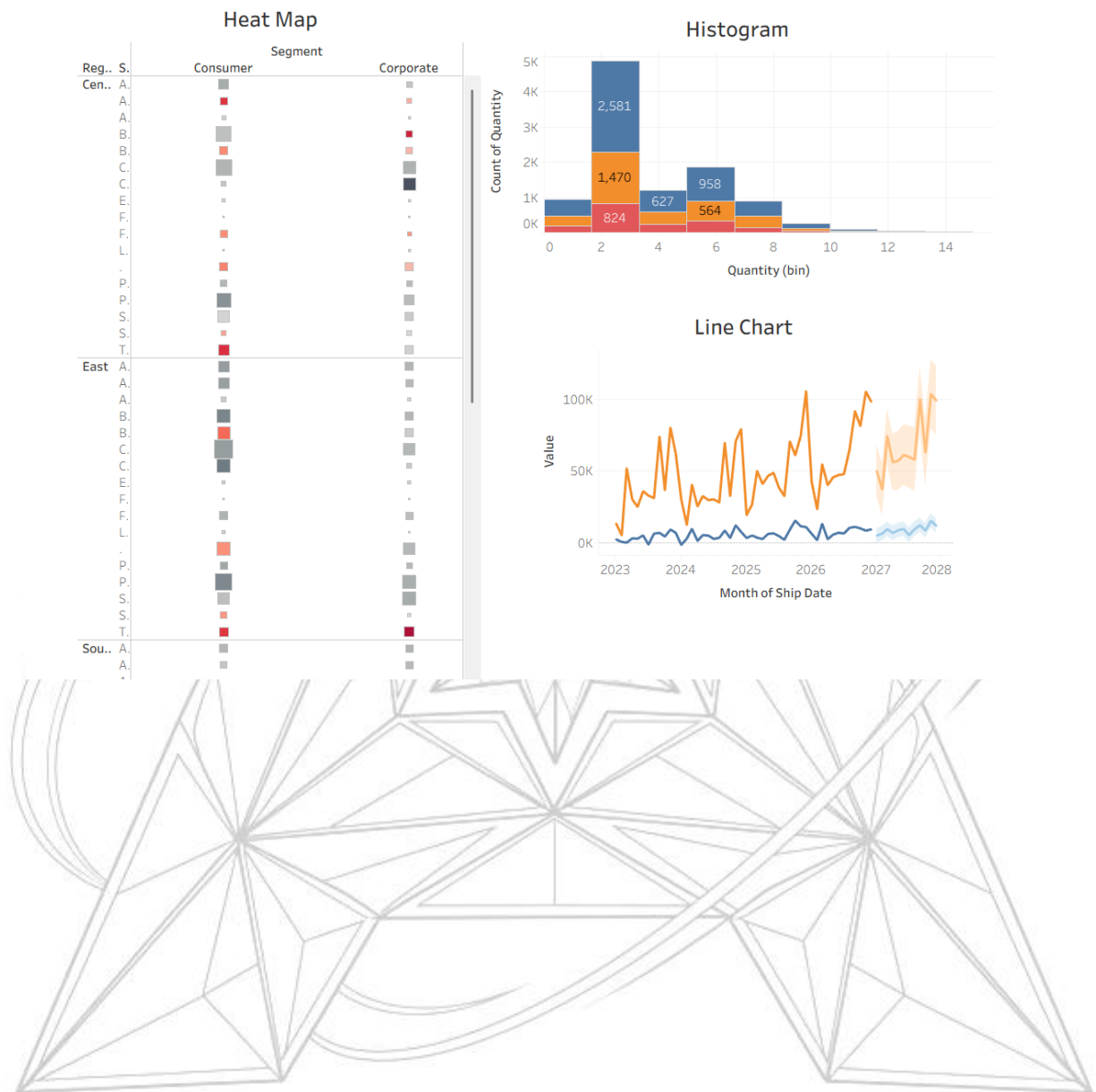


DASHBOARD 02

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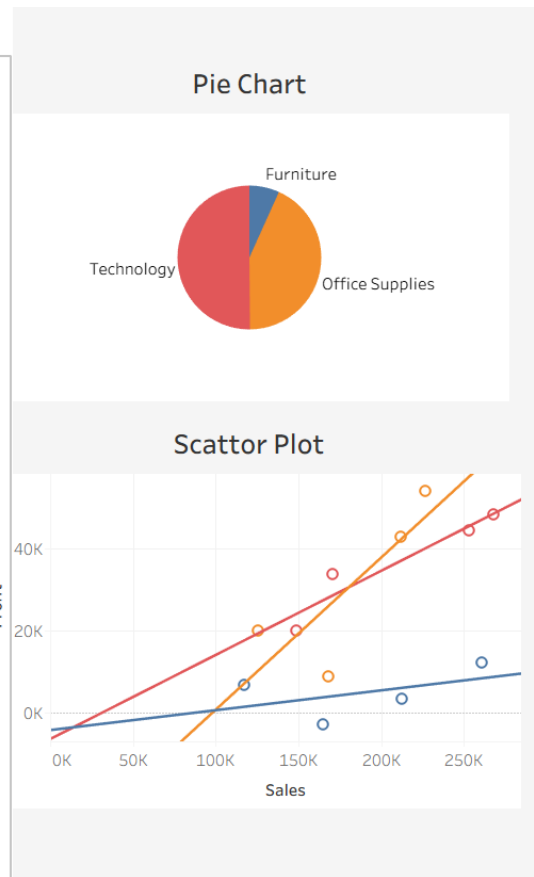
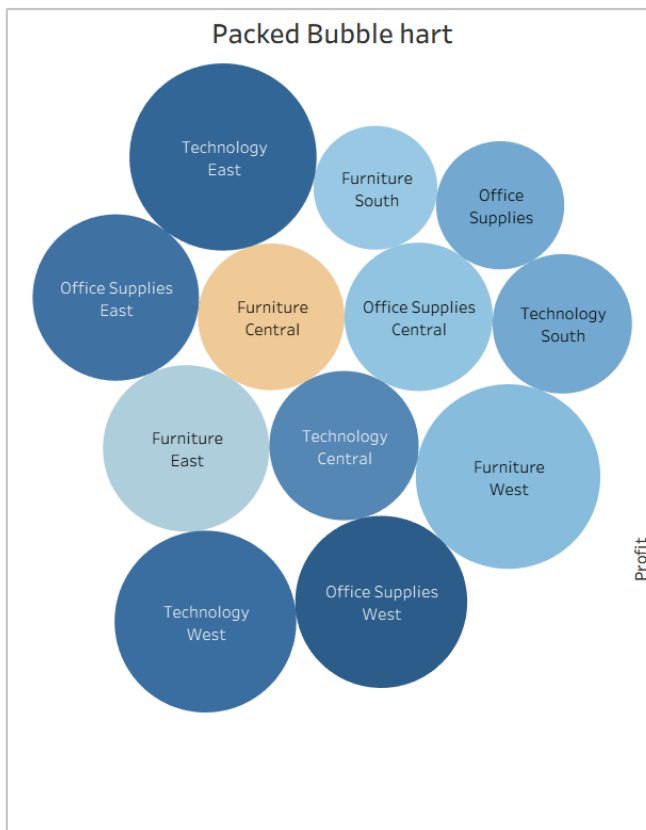




DASHBOARD 03

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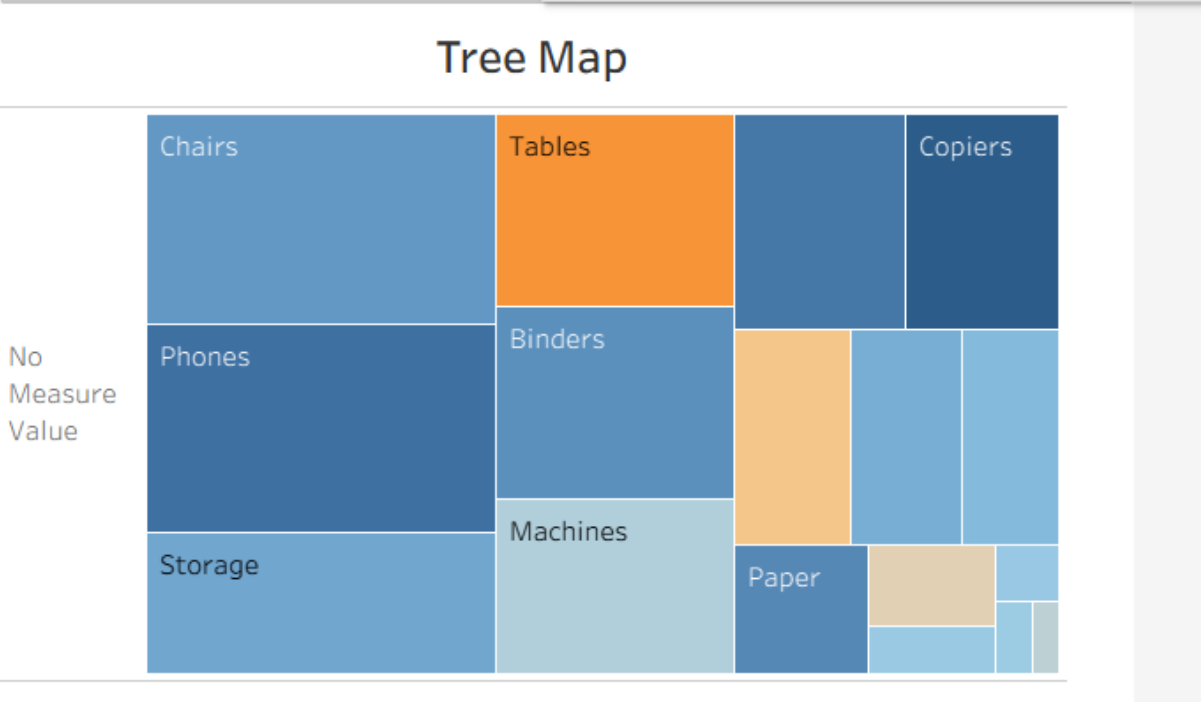


DASHBOARD 04

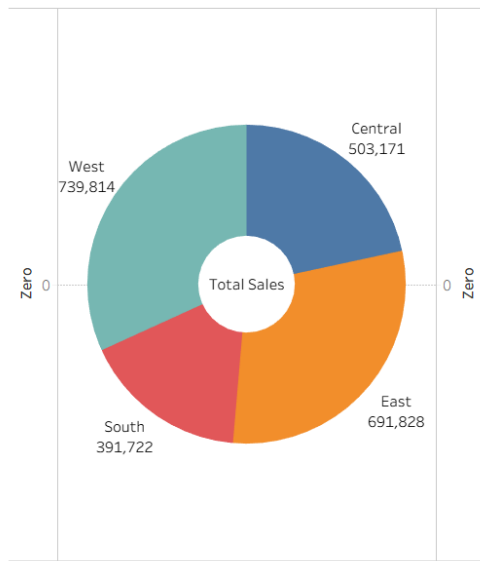
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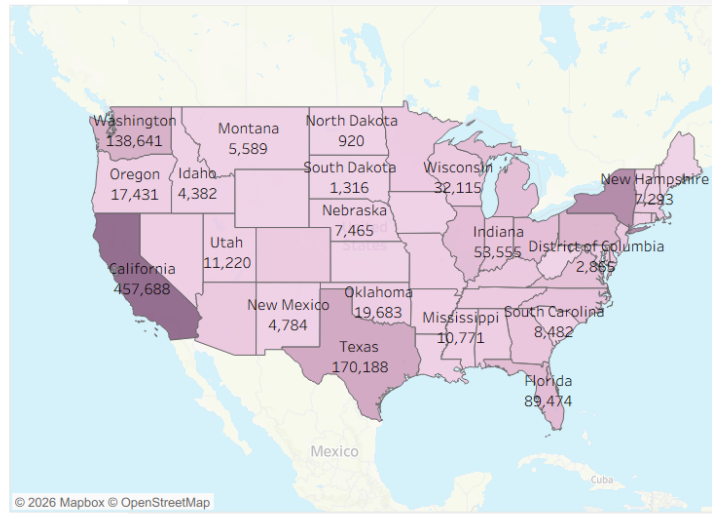
Text Table with Percentage					
Region	Sub-Catego..	Order Date			
		2023	2024	2025	2026
Central	Accessories	4.27%	7.58%	7.27%	7.39%
	Appliances	3.52%	4.84%	4.05%	6.04%
	Art	0.79%	1.10%	1.02%	1.55%
	Binders	15.28%	5.73%	10.19%	14.31%
	Bookcases	1.77%	8.07%	5.64%	3.81%
	Chairs	19.99%	17.41%	15.74%	16.15%
	Copiers	3.15%	12.45%	11.78%	2.49%
	Envelopes	1.54%	0.85%	0.65%	0.81%
	Fasteners	0.12%	0.09%	0.17%	0.22%
	Furnishings	2.44%	2.46%	3.45%	3.44%
	Labels	1.01%	0.30%	0.34%	0.40%
	Machines	15.69%	1.00%	1.78%	1.88%
	Paper	2.26%	3.00%	2.00%	2.00%
	Phones	9.56%	18.00%	18.00%	18.00%
	Storage	10.68%	8.00%	8.00%	8.00%
	Supplies	0.42%	0.00%	0.00%	0.00%
	Tables	7.50%	6.00%	6.00%	6.00%
		Region: Central			
		Sub-Category: Labels			
		Year of Order Date: 2024			
		% of Total Sales along Pane (Down): 0.30%			



Donut Chart



Map Chart



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